PINION GEAR/RING GEAR (Continued)

CAUTION: Never loosen the pinion nut to decrease pinion bearing rotating torque and never exceed specified preload torque. Failure to heed caution may result in damage.

(9) Using Spanner Wrench 6958 and torque wrench set at 678 N·m (500 ft. lbs.) (Fig. 76). Slowly tighten the nut in 6.8 N·m (5 ft. lb.) increments until the rotating torque is achieved. Measure rotating torque frequently to avoid over crushing the collapsible spacer (Fig. 77).

CAUTION: If more than 678 N·m (500 ft. lbs.) torque is required to crush the collapsible spacer, the spacer is defective and must be replaced. Failure to heed caution may result in damage.

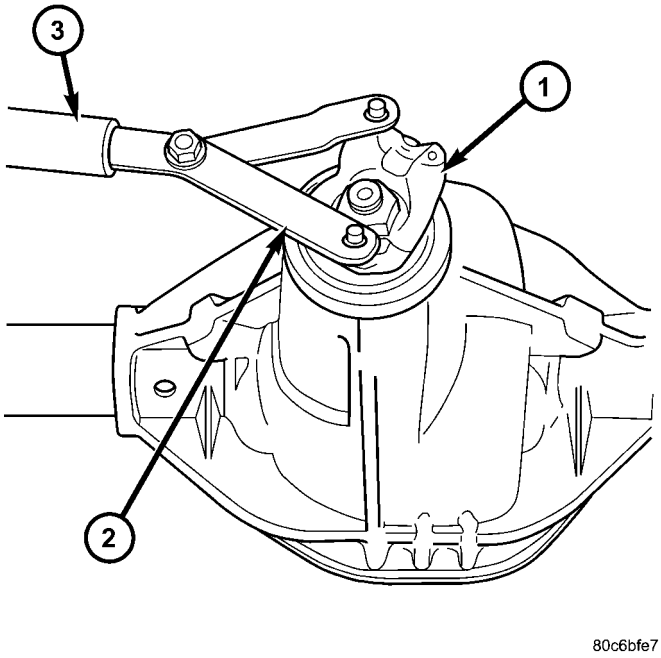

Fig. 76 YOKE SPANNER WRENCH

1 - PINION YOKE
2 - SPANNER WRENCH
3 - PIPE

(10) Check bearing rotating torque with an inch pound torque wrench (Fig. 77). The torque necessary to rotate the pinion should be:

- Original Bearings: 1.13 to 2.26 N·m (10 to 20 in. lbs.).
- New Bearings: 1.7 to 3.4 N·m (15 to 30 in. lbs.).

(11) Invert the differential case and start two ring gear bolts. This will provide case-to-ring gear bolt hole alignment.

(12) Invert the differential case in the vise and install **new** ring gear bolts and alternately tighten to 108 N·m (80 ft. lbs.) (Fig. 78).

CAUTION: Never reuse ring gear bolts. Failure to heed caution may result in damage.

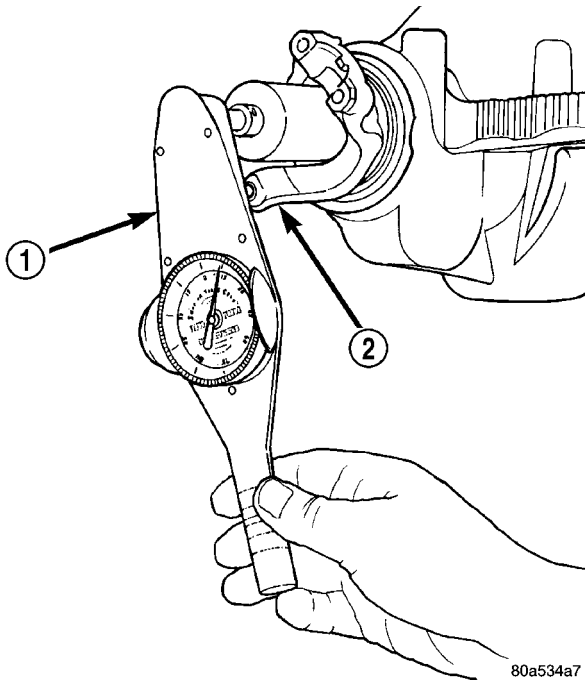

Fig. 77 PINION ROTATION TORQUE

1 - TORQUE WRENCH
2 - PINION YOKE

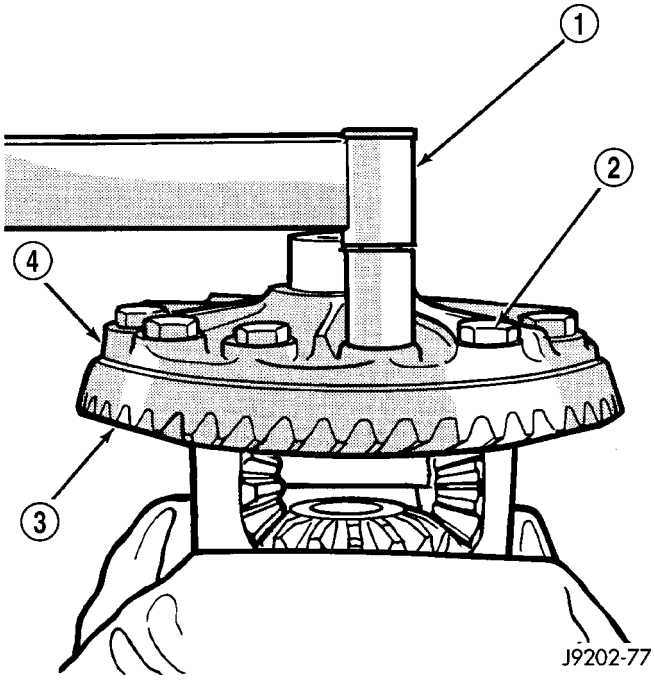

Fig. 78 RING GEAR BOLT

1 - TORQUE WRENCH
2 - RING GEAR BOLT
3 - RING GEAR
4 - CASE

(13) Install differential and verify differential bearing preload, gear mesh and contact pattern.